

Shubham Sinha

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Technical Skills

Core AI & ML:	LLMs, Agentic AI, RAG, Text-to-SQL, Supervised Learning, Model Evaluation
Optimization:	Gurobi, Mathematical Modeling, Heuristics, Linear Programming
Programming & Ops:	Python, SQL, REST APIs, Docker, Flask, Git, AWS (SageMaker, Lambda)
Data Platforms:	GCP, Azure, PostgreSQL, NumPy, Pandas, Folium

Work Experience

Deloitte USI, Hyderabad

Aug 2023 – Present

Data Scientist – AI & Advanced Analytics

US State Client – Enterprise Agentic AI (Retirement Plan)

- Developed 2 agents for a public-facing bot, creating a scalable multi-agent system (MCP ready) supporting **1,500+ concurrent users** and **28,500+ queries** in 5 minutes.
- Engineered production-grade software guardrails, including **400+ automated safety tests**, ensuring system reliability.
- Developed an intent recognition agent to classify queries across 10 categories, optimizing document filtering and response tone.
- Implemented high-performance cache retrieval mechanisms, reducing **average response latency to 1.1 seconds**.
- Impact:** Technical excellence contributed to the client winning the RFP to retain the project for the US state government.

Metropolitan Planning Organization – Geospatial Data Engineering & AI

- Led a development team of 2**, operating with high autonomy and reporting directly to Senior Leadership.
- Curated 100+ geospatial datasets, transforming raw (Silver) inputs into analytical-grade (Diamond) quality.
- Automated reporting pipelines using Python and Folium, slashing interactive analytics generation time to **2 minutes**.
- Global Recognition:** Solution featured in **BBC** and **The Wall Street Journal** for urban planning innovation.

US State Transportation Client – Large-Scale GenAI Data Platform

- Engineered a data pipeline to conflate **4+ Petabytes** of transportation data, achieving **95% accuracy** across 10+ use cases.
- Delivered the firm's first GenAI transportation prototype, positioning Deloitte as a frontrunner for **\$100M** contracts.
- Optimized incident report processing for 4,000+ reports, achieving processing speeds of **1-2 seconds per report**.

Udaan, Bangalore

May 2022 – Dec 2022

Data Science Intern (Optimization)

- Optimized Transshipment Route formulation using mathematical modeling and Gurobi, resulting in operational savings of **≈10 Lakhs per month**.
- Reduced computational time by **50–68%** for complex routing (30x30 matrix) by integrating heuristic approaches with exact solvers to optimize hop constraints.
- Enhanced system scalability and logistics performance by refactoring code for the Transshipment Route Problem.

Education

IIT BHU (Varanasi)

2021 – 2023

M.Tech. in Decision Science and Engineering — CGPA: 9.23/10

Relevant Coursework: Machine Learning, Optimization, Algorithms

BIT Sindri, Dhanbad

2017 – 2021

B.Tech. in Production Engineering — CGPA: 8.36/10

Relevant Coursework: Operations Research, Applied Mathematics

Projects & Leadership

- Weld Quality Monitoring (Deep Learning):** Built a CNN-based real-time quality prediction system for automotive assembly, achieving **85% accuracy**.
- Placement Coordinator (IIT BHU):** Served as Student Representative for the Batch of 2023, managing recruitment.
- Course Coordinator:** Managed “Data-Driven Supply Chain Transformation” course, coordinating logistics for **1,600+ applicants**.

Awards & Certifications

- Performance Awards:** Recipient of 2 Outstanding Awards, 1 Applause Award, and 1 Spot Award.
- Certifications:** AWS Certified Machine Learning – Specialty, Azure AI Engineer Associate.